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Palaeomagnetic results from the Early Permian Copacabana Group, southern Peru: Implication for Pangaea palaeogeography

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Abstract

Samples collected from folded carbonate rocks of the Early Permian Copacabana Group exposed in the Peruvian Subandean Zone have been subjected to detailed palaeomagnetic analysis. Thermal demagnetisation of most samples yield stable high unblocking temperature directions dominantly carried by titanomagnetite minerals. This remanence, identified in 32 samples (43 specimens), is exclusively of reverse polarity consistent with the Permian–Carboniferous Reversal Superchron (PCRS). The overall directions pass the fold test at the 99% confidence level and are considered as being a pre-folding remanence acquired in Early Permian times. The Copacabana Group yields an overall mean direction of $D=166^\circ$, $I=+49^\circ$ ($\alpha_{95}=4.5^\circ$, $k=131.5$, $N=9$ sites) in stratigraphic coordinates and a corresponding palaeosouth pole position situated at $\lambda=68^\circ$S, $\phi=321^\circ$E ($A_{95}=5.2^\circ$, $K=100$). Combining this pole with the coeval high quality data from South America, Africa and Australia results in a mean pole for Gondwana situated at $\lambda=34.4^\circ$S, $\phi=065.6^\circ$E ($A_{95}=4.9^\circ$, $K=73.6$, $N=13$ studies) in African coordinates. This pole position supports a Pangaea B palaeogeography in Early Permian times. In contrast, the combined pole for Gondwana diverges from the coeval Laurasian mean pole when assuming the Pangaea A-type configuration. Poor quality of the Gondwana dataset and inclination shallowing in sediments seem to play no role in the misfit between the Permian–Triassic poles from Gondwana and Laurasia in Pangaea A reconstruction.
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1. Introduction

The geometry of Pangaea, a supercontinent which formed by amalgamation of Gondwana, Laurasia, Siberia and other smaller continental fragments and thought to have existed throughout Permian and Triassic times, remains a matter of discussion. This is because palaeogeographic reconstructions based on Permian palaeomagnetic data do not agree with the Jurassic Pangaea type A configuration proposed by Bullard et al. (1965), which is a well-accepted starting point for the break-up of Pangaea and subsequent opening of the Atlantic Ocean. This disagreement with the Pangaea A configuration, which places South America and Africa (Gondwana) facing North America and Europe (Laurasia), is demonstrated by the misfit of the Gondwanan and Laurasian palaeomagnetic pole positions when adopting this palaeogeography in Permian times. Similarly, palaeogeographic reconstruction

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based on Permian palaeomagnetic data provides unacceptable overlap between the Gondwanan and the Laurasian continental margins when South America is placed facing North America such as in Pangaea A. Alternative reconstructions known as Pangaea B (Morel and Irving, 1981) and Pangaea C (Smith et al., 1981) have been proposed in order to overcome this overlap. Gondwana is placed farther to the east, with South America facing Europe and Africa facing Eurasia, in types B and C configurations. Although palaeomagnetic data provide a better match when adopting such relative position between Gondwana and Laurasia, Pangaea B and C are questioned by many authors because it implies an approximately 3500 km east–west relative translation (Pangaea B) or more (Pangaea C) during the Permian–Triassic in order to reach Pangaea A in Jurassic times. As no straightforward geological evidence for such a large-scale translation event has been found, it has been suggested that the mismatch of the palaeomagnetic data in Pangaea A configuration is the result of the poor quality of the existing palaeomagnetic dataset for Gondwana and/or inclination shallowing in sediments (Van der Voo, 1993; Rochette and Vandamme, 2001). Another hypothesis suggesting that this mismatch is the result of significant contributions of non-dipole components to the Earth's geomagnetic field during the Late Palaeozoic–Early Mesozoic has also been proposed by Van der Voo and Torsvik (2001) and Torsvik and Van der Voo (2002). Conversely, some authors (e.g. Muttoni et al., 1996; Torcq et al., 1997; Muttoni et al., 2003) maintain that high quality data from Gondwana are trustworthy with no evidence for inclination shallowing error or significant non-dipole field contributions. These authors regard Pangaea as an evolving supercontinent that developed from and Early Permian B-type to an Early Jurassic A-type configurations. Different timings of the E–W relative motion of Gondwana with respect to Laurasia responsible for this transformation, however, have been proposed. On one hand, Torcq et al. (1997) proposed that the westward translation of Gondwana occurred in Triassic times. Compatibility with the general extensional pattern in the tethyan region and good agreement between the Late Triassic palaeomagnetic poles from Gondwana and Laurasia were reported to sustain this model. Muttoni et al. (1996, 2003), on the other hand, based on palaeomagnetic data and some geological supports, propose that the transformation from type B-type to A-type configurations was constrained within the Permian times and Pangaea has already reach an A-type configuration in Late Permian. To help resolve these problems, a detailed investigation of Early Permian rock units from stable southern Peru has been carried out in an effort to contribute high-quality data to this contest.

2. Geological setting

The sampling area for the present palaeomagnetic investigation is situated within the Peruvian Subandean Zone in the region north of Cuzco, which is delimited by the Madre de Dios-Beni foreland basin to the east and by the Eastern Cordillera to the west (Fig. 1). This segment of the Andes constitutes a deflexion zone (Ambacay deflexion) that separates the N–S trending northern segment and the NW–SE trending southern segment of the Peruvian Andes. While general Andean structures in the southern Peruvian Subandean Zone are dominated by NW–SE trending northeast-verging fold and thrust belt (Gil et al., 1999; Jaillard et al., 2000),

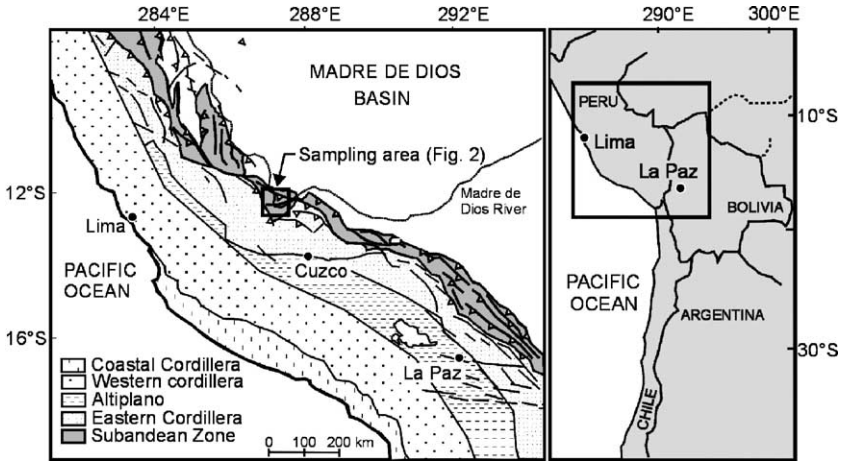


Fig. 1. Simplified morphostructural map of the Peruvian Andes showing location of the sampling area (box).

deformational features in the sampling area are dominated by E–W trending structures (De la Cruz et al., 1998), with large synclines of Tertiary sedimentary cover to the north, and thrusted Ordovician to Early Permian sediments to the south (Fig. 2). These structures were developed during the Cenozoic Andean orogeny, namely during the Late Eocene Incaic and the Middle Miocene Quechua phases, and during the Miocene–Pliocene compression (De la Cruz et al., 1998). The Incaic compressional phase was responsible for the E–W and NW–SE trending folds and thrusts in the area. Folding of the Paleogene and Neogene sediments is attributed to NE–SW and N–S shortening during the Quechua phase. The Miocene–Pliocene W–E compressional regime resulted bending of the Paleozoic and Paleogene–Neogene sediments in the area northeast of Pongo de Mainique (De la Cruz et al., 1998).

Paleozoic outcrop in the area include Ordovician and Devonian marine sediments interpreted to have deposited in retroarc-foreland basin that developed along the western margin of South America (Jaillard et al., 2000). The overlying Carboniferous to Permian sediments (Cuevo Supersequence) are thought to have deposited during regional extensional tectonic and basin subsidence along this continental margin (Sempere, 1995). In Peru, Permo-Carboniferous sedimentary sequence include fluvial and deltaic clastic deposits (Ambo Group) at the base, and detritic sediments overlain by stable marine platform carbonates of Late Carboniferous–Early Permian age (Tarma and Copacabana Groups) on top. In the Subandean Zone, the Copacabana Group is conformably overlain by Early to Middle Permian calcareous sandstones of the Ene Formation or its equivalent. Fossils found at Pongo de Mainique are interpreted as indicating an Early to Middle Permian (Artinskian-Kungarian) age for the Ene Formation (De la Cruz et al., 1998). Stratigraphic correlation made by Sempere et al. (2002), however, suggests that the Ene Formation and its equivalent (Ampay Formation in Peru, Viticua Formation in Bolivia) may have deposited throughout the Middle Triassic times.

Generally, the Permo-Carboniferous sequence (Cuevo Supersequence) in the central Andes is bounded by surface discontinuities, which are associated with angular unconformity and folding of the underlying sediments in some places. These surfaces have been interpreted as being unconformities related to regional deformational events during the mid-Carboniferous eohercynian (at the base) and the Late Permian late hercynian (on top) (Dalmayrac et al., 1980). Recent studies carried out by Díaz-Martinez et al. (2000) and Sempere et al. (2002), how-

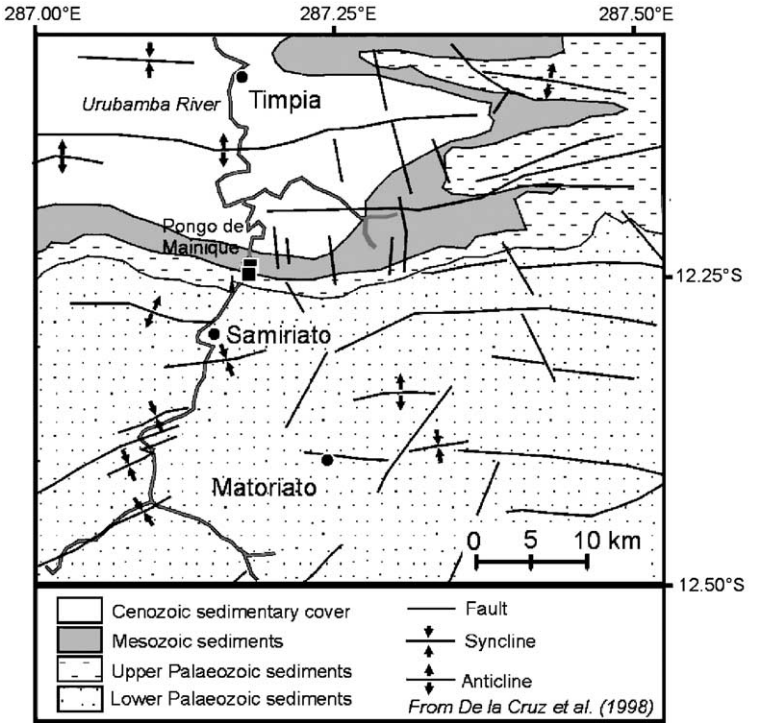


Fig. 2. Simplified geological map of the study area (box in Fig. 1). Sampling location at S_{lat}=12.2°S/S_{long}=287.2°E (squares) for Copacabana Group (MCO) and La Chonta Formation (MCR) are shown.

ever, provided evidence suggesting that these deformations are only localised and simultaneously thick conformable sediments were deposited in many places. According to Sempere et al. (2002), this differing stratigraphy relationships strongly suggest a back-arc transcurrent extensional setting for the Central Andes, in which transtensional and transpressional regime prevailed concurrently during the Carboniferous and throughout the Permian.

During the Late Permian to Jurassic times, continuous extensional regime associated with the break-up of Pangaea resulted in NW–SE partial rifting in the area west of the Subandean Zone (Jaillard et al., 2000; Ramos and Aleman, 2000; Sempere et al., 2002). Sediments accumulated within the rift system include the syn-rift Late Permian–Early Triassic red-beds and volcanics of the Mitu Group, which overlies the Copacabana Group in the Eastern cordillera and Altiplano areas, and the Triassic and Jurassic post-rift sediments (Carlotto, 2002; Sempere et al., 2002). Conversely, at the sampling location (Pongo de Mainique) and, in many areas of the Subandean Andes, the Mitu Group is absent and Permian sediments (Copacabana Group/Ene Formation) are directly overlain by Late Cretaceous sediments (La Chonta and Vivian Formations). This discontinuity indicates that the sampling area was uplifted and formed a high topography during the Triassic to Early Cretaceous times (Gil et al., 1999; Carlotto, 2002) and became site of erosion and/or depositional hiatus (De la Cruz et al., 1998). Late Permian transcurrent rift system, in which transtensional accommodation would have been separated by transpressional uplift segments (Sempere et al., 2002), is considered responsible for the palaeogeography of the Central Andes during these times.

From Late Cretaceous onward, deformation in the Subandean back-arc areas are related to the thrusting of the Andean chain upon the stable south America plate and are generally characterised by compressional shortening yielding east-verging fold and thrust belts. Deformational style and the amount of shortening are, however, highly variable along the Andean chain (Gil et al., 1999). Andean structures in the sampling area are discussed above.

3. Sampling details

Since our sampling in southern Peru was undertaken in an attempt to provide continuous palaeomagnetic data from Palaeozoic rocks, we also collected samples from the Cabanillas, the Ambo and the Tarma Groups in addition to the Copacabana Group. Although unweathered rocks were targeted during sampling, the 205 samples drilled in 38 sites from Cabanillas, Ambo and Tarma Groups did not provide reliable palaeomagnetic data.

Samples for the Copacabana Group were collected at Pongo de Mainique, along the Urubamba River (Fig. 2), where the group consists of an approximately 800 m thick carbonate sequence comprising flora and fauna-rich grey micritic, nodular, detritic limestones and calcareous sandstones. The Copacabana Group is dated as Upper Wolfcampian (Asselian-Sakmarian) based on palaeontological and palynological studies (De la Cruz et al., 1998; Quiñones, 1990; Azcuy et al., 1992). Stratigraphically, the Copacabana Group conformably overlies the Tarma Group and is, in turn, overlain by the Ene Formation (=Rio Tambo Formation of De la Cruz et al., 1998). Both the Tarma Group and the Ene Formation are biostratigraphically well-dated as being Late Carboniferous and end Early to Middle Permian in age, respectively (De la Cruz et al., 1998 and references therein). Samples were collected from steeply east dipping beds (6 sites) and from both limbs of an approximately 20 m folded carbonate sequence (5 sites), with fold axis dipping 15° towards west (Fig. 3). Altogether, 56 cores drilled in nodular limestone, detritic limestone, and calcareous sandstone beds were investigated. Additional Late Cretaceous samples (La Chonta Formation) were collected at the same location to provide tectonic control. The La Chonta Formation comprises a fossiliferous marine carbonate succession, palaeontologically dated as Coniacian–Santonian (83–89 Ma; De la Cruz et al., 1998 and references therein). Fourteen core samples were drilled in an approximately 3 m thick subvertical limestone bed. No further sampling for this unit was possible due to limited outcrop exposure.

4. Palaeomagnetic methods

Palaeomagnetic measurements were carried out at the palaeomagnetic laboratory of the Geophysics section in the Ludwig-Maximilians-Universität, Munich, Germany. Each drill core was cut into standard cylindrical specimens, which were subject to either stepwise thermal or alternating field demagnetisation up to 580 °C/0.2 T using Schoenstedt furnaces and the 2G AF demagnetising system. The direction of the initial natural remanent magnetisation (NRM) and the direction of the remaining magnetisation after each demagnetisation step were measured with 2G-cryogenic magnetometer (DC-SQUID). Demagnetisation results for individual specimens were analysed using both orthogonal (Zijderveld, 1967) and equal area stereo-

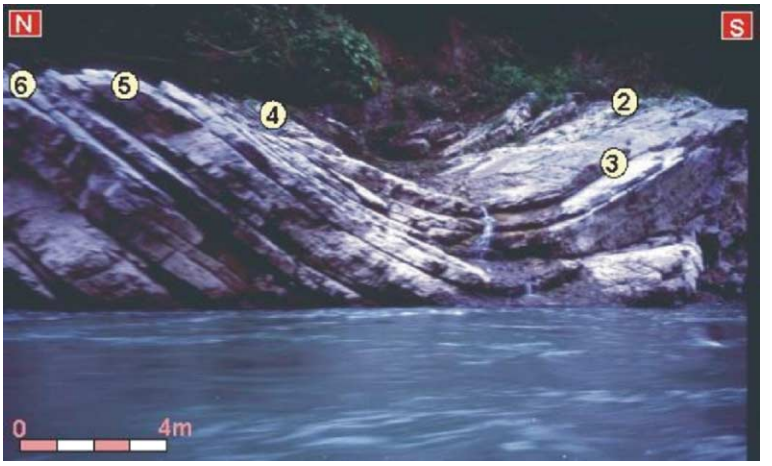


Fig. 3. Photograph of the folded carbonate unit sampled at Pongo de Mainique. Sampling sites (numbers 2 to 6) are shown. The plunge of the fold axis (15° toward NW) was taken into consideration when correcting palaeomagnetic data from these sites. Sampling sites MCO1 and MCO7-11 are situated south and north of this folded unit, respectively.

graphic projection. Linear trajectories of vector endpoints in orthogonal projection were identified by eye and magnetic directions were calculated using principal component analysis (Kirschvink, 1980). The mean directions were calculated using statistics after Fisher (1953).

5. Palaeomagnetic results and interpretations

The carbonate rocks from the Copacabana Group are generally weakly magnetised, with initial NRM intensity ranging between 0.04 mA/m and 0.5 mA/m. Stepwise demagnetisations of the specimens cut from 56 samples yield two distinct magnetic directions, a low unblocking temperature component (A), which is completely removed after heating to 270 °C, and the higher unblocking temperature component (B), which is stable up to 480–520 °C. Both directions are well defined in orthogonal projection of results (Fig. 4). The A direction is generally defined between 100 °C and 250 °C (AF: 5–20 mT) demagnetisation steps. Individual directions were calculated from linear trajectory of vector endpoints that do not converge towards the origin in orthogonal projection and yield an overall site-mean direction of $D=006.3°$, $I=-20.1°$ ($\alpha_{95}=5.9°$, $k=80.6$, $N=8$ sites, $n=38$ samples) in-situ. Directional dispersion into two distinct groups after bedding correction (Fig. 5) clearly indicates that component A was acquired posterior to folding of the rocks. As the in situ mean direction plots at the vicinity of the expected current dipole field direction at the sampling location, this magnetisation is likely acquired under the present day geomagnetic field.

The high unblocking temperature component B is identified in 32 samples from 9 palaeomagnetic sampling sites. It is well defined by linear decay of the magnetisation, with data points showing minor directional changes and reach, or are clearly directed toward the origin in orthogonal projection (Fig. 4). Individual directions were calculated from at least the last five demagnetisation steps above 250 °C (AF=20 mT). The majority of the specimens are totally demagnetised between 480 °C and 520 °C (AF=90 mT), suggesting a remanent magnetisation carried predominantly by titanomagnetite. In contrast to component A, the sample and site level directions for remanence B are distributed into two distinct groups in in-situ coordinates, but are well clustered with southerly declinations with positive inclination values after bedding correction (Fig. 5). The overall site-mean direction is $D=158.5°$, $I=+00.2°$ ($\alpha_{95}=94.4°$, $k=2.5$) in-situ, and $D=166.1°$, $I=+48.9°$ ($\alpha_{95}=4.5°$, $k=131.5$, $N=9$ sites) after bedding correction (Table 1). Component B passes the fold test of McElhinny (1964, K-ratio=82.03) and that of McFadden (1990), both at the 99% level of confidence. Stepwise unfolding shows maximum grouping of the site-mean directions at $101\pm8\%$ ($k=236.3$, $\alpha_{95}=3.4°$; Enkin, 2003). Component B was, therefore, acquired prior to folding of the unit in Permian times as a result of either the late Hercynian compressional event (De la Cruz et al., 1998; Jaillard et al., 2000), or the transpressional/transtensional regime that prevailed until the Late Permian times (Sempere et al., 2002). Hence, B remanence was most likely acquired during, or shortly after deposition of the Copacabana Group. The overall mean palaeosouth pole calculated from the site-level

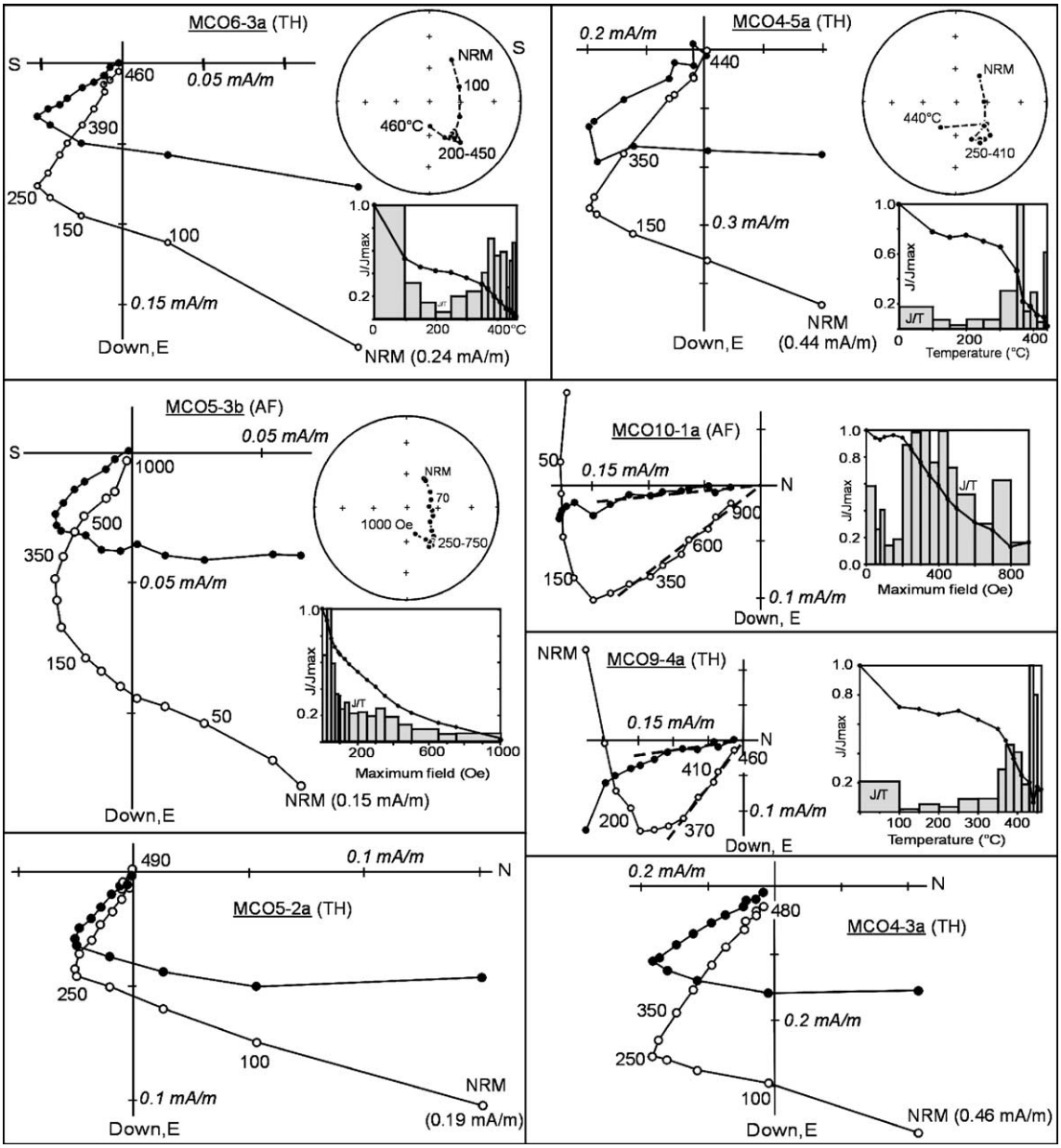


Fig. 4. Examples of stepwise thermal and alternating field demagnetisation results from Copacabana Group. Data points in orthogonal projections are shown in bedding-corrected coordinates. The high unblocking temperature component B is defined within the 250 °C to 480 °C temperature range. Open/close circles in orthogonal projections represent data points in the horizontal/vertical plane. NRMs represent initial intensity of remanence for each sample.

VGPs (Table 1) is situated at $\lambda = 68.2°$S, $\phi = 321.3°$E ($A_{95} = 5.2°$, $K = 99.8$, $N = 9$ sites) in south American coordinates, and $\lambda = 40.6°$S, $\phi = 051.3°$E when rotated into African coordinates according to the reconstruction parameters of Lawver and Scotese (1987).

The overlying Late Cretaceous La Chonta Formation, which was sampled to provide additional tectonic control for the Permian results, has NRM intensities ranging between 3.1 mA/m and 4.6 mA/m. Thermal demagnetisation up to 690 °C of all samples yield a well-defined stable magnetisation. Individual sample directions were calculated from linear trajectories of vector components between 250 °C to 650 °C, clearly converging towards the origin in orthogonal projection (Fig. 6a). Unblocking temperatures above 650 °C indicates haematite as the dominant remanence carriers. The overall results from

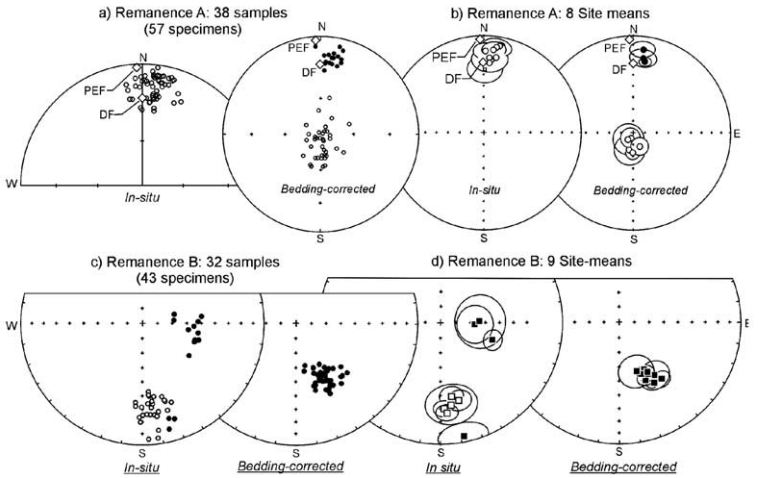


Fig. 5. Equal area stereographic projection of the palaeomagnetic data from the Copacabana Group. (a, b) Sample and site-mean directions for the present day overprint (component A). The expected present magnetic field (PEF) and the present magnetic dipole field (DF) at the sampling area are shown for comparison. (c, d) Sample and site-mean directions for the stable remanence B. Site-mean directions are shown along with their corresponding confidence circles (α_{95}). Closed/open symbols represent positive/negative inclination.

La Chonta Formation (LCF) yield a mean of $D=183^\circ$, $I=-47.4^\circ$ ($\alpha_{95}=2.5^\circ$, $k=259$, $n=14$ samples) in-situ, and $D=183.5^\circ$, $I=+36.6^\circ$ ($\alpha_{95}=2.5^\circ$, $k=259.3$) after bedding correction. No change in directional grouping was expected as all samples were collected from unit of uniform structural attitude.

Due to limited outcrop, no field test could be carried out to constrain the relative age of the LCF magnetisation. As all data are from an approximately 3 m thick limestone bed, it may represent a direction that has not averaged the secular variation of the Earth magnetic field. The corresponding pole could, therefore, represent a Late Cretaceous virtual geomagnetic pole. However, it is worth to discuss all possible options for the results from La Chonta Formation and their tectonic implications.

In any case, the in-situ LCF direction deviates from all expected Late Cretaceous and younger directions at the sampling area recalculated from the reasonably well-defined reference poles for South America (Neogene: $D=178.0^\circ$, $I=31.1^\circ$, $\alpha_{95}=5^\circ$; Paleogene: $D=183.7^\circ$, $I=38.3^\circ$, $\alpha_{95}=12^\circ$; Late Cretaceous: $D=172.9^\circ$, $I=22.2^\circ$, $\alpha_{95}=5^\circ$; Randall, 1998). If one assume that LCF was acquired posterior to the rock deformation but deviates from the reference directions due to tectonic rotation, this implies a large counterclockwise rotation (175–181°) and significant latitudinal displacement (9–16°) of the sampling location relative to stable South America. As this is incompatible with the geology of the sampling area, LCF direction is unlikely a post-folding Neogene or Paleogene remagnetisation. On the other hand, one may also consider that LCF is either a primary Late Cretaceous direction rotated during the Andean deformation. Comparison with the expected Late Cretaceous direction at sampling location, however, shows that the bedding-corrected LCF direction has a steeper inclination, implying an irreconcilable latitudinal inconsistency ($10.6\pm4^\circ$) of the sampling area relative to stable South America. Since local rotation about vertical axis should only affect the declination component of magnetisation, this discrepancy rule out the possibility for LCF direction as being a rotated Late Cretaceous magnetisation. Conversely, an assumption that the result from La Chonta Formation may represent a pre-folding secondary direction is supported by the coincidence of the palaeomagnetic south pole ($\lambda=81.1^\circ$S, $\phi=264.8^\circ$E, dp=2.9°, dm=1.7°) corresponding to the LCF bedding-corrected with the proposed Palaeogene reference directions for stable South America (Fig. 6c) compiled by Randall (1998, 24–66 Ma), Beck (1998, 30 Ma), Van der Voo (1993, 30–67 Ma) and Roperch and Carlier (1992, 30 Ma). This suggests that LCF may have acquired in Paleocene–Oligocene (24–66 Ma) times and the sampling area was stable since then. As development of the thrust belt in the Subandean Zone occurred during the Late Miocene times (Baby et al., 1999) and the deformation of the la Chonta Formation is attributed to the N–S shortening during the Middle Miocene Quechua phase (De la Cruz et al., 1998), the only possibility that satisfy both the geology of the area and the presented palaeomagnetic data is to regard LCF magnetisation a pre-folding overprint acquired between the Paleocene and

Table 1
Site-mean directions for magnetic components from Copacabana Group and La Chonta Formation

Site	*n*	Dg (°)	Ig (°)	α_{95} (°)	*k*	Ds (°)	Is (°)	α_{95} (°)	*k*
Copacabana Group—component A									
MCO1	4	1.2	−34.8	13,5	25.7	213.1	−83.2	13,5	26
MCO2[a]	3	8.5	−24.6	18.8	44.2	365.5	−72.6	18.8	44
MCO3[a]	4	5.5	−37.9	21.8	18.8	27.3	−79.3	21.8	19
MCO4	4	7.0	−14.9	7.6	102.7	7.1	14.4	7.6	103
MCO5	3	8.8	−11.9	9.3	176.5	8.1	23.6	9.3	177
MCO6	5	10.9	−10.0	3.4	233.3	9.3	26.2	3.4	233
MCO7	5	4.9	−25.0	12.7	28.7	189.7	−75.1	12.7	29
MCO8[a]	2	3.8	−17.1	25.5	98.1	200.3	−81.7	25.5	98
MCO9	5	9.8	−23.2	7.4	57.2	153.9	−77.1	7.4	57
MCO10	4	1.3	−16.7	8.2	68.2	202.5	−85.0	8.2	68
MCO11	5	5.3	−25.9	5.7	96.5	179.5	−73.1	5.7	97
Site-mean direction		Dg (°)	Ig (°)	α_{95} (°)	*k*	Ds (°)	Is (°)	α_{95} (°)	*k*
		6.3	−20.3	6.2	80.6	9.6	−56.8	49.9	2.0
Palaeopole: λ=83.6°S/ϕ=183.6°E (A_{95}=4.1, K=185.9)									
Copacabana Group—component B									
MCO3	5	168.8	4.5	10.5	29.9	161.3	47.4	4.4	127
MCO4	3	108.6	52.8	5.4	136.4	149.9	49.7	1.8	614
MCO5	3	88.5	63.2	11.4	50.8	162.3	56.3	9.1	33
MCO6	3	92.8	66.9	7.6	67.5	162.5	54.1	9.0	144
MCO7	3	167.1	−34.6	5.7	121.1	165.0	42.0	7.7	1001
MCO8	4	172.7	−32.2	10.1	65.0	170.3	45.5	9.9	87
MCO9	3	177.8	−29.7	5.1	149.4	176.3	48.0	6.3	150
MCO10	4	176.1	−26.6	5.5	110.3	173.3	50.8	6.4	110
MCO11	4	171.9	−38.6	4.8	168.1	172.3	43.8	5.9	168
Site-mean direction		Dg (°)	Ig (°)	α_{95} (°)	*k*	Ds (°)	Is (°)	α_{95} (°)	*k*
		158.5	0.2	94.4	2.5	166.1	48.9	4.5	131.5
Palaeopole: λ=68.2°S/ϕ=321.3°E (A_{95}=5.2, K=99.8)									

La Chonta Formation

Mean (n=14 samples)	Dg (°)	Ig (°)	Ds (°)	Is (°)	α_{95} (°)	*k*
Direction	183.0	−47.4	183.5	36.6	2.5	259.3
Pole: λ=81.1°S/ϕ=264.8°E (dp=2.9°, dm=1.7)						

n: number of samples used for calculation of site mean; Dg/Ig and Ds/Is: declination/inclination (in degrees) in geographic (g) and stratigraphic (s) coordinates; α_{95} (°): semi-angle of the cone of 95% confidence; *k*: precision parameter after Fisher (1953); λ (°S) and ϕ (°E): latitude and longitude position of the palaeosouth poles calculated from site-level VGPs; dp/dm: axes of 95% oval of confidence about the palaeomagnetic pole; A_{95} (°): semi-angle of the 95% confidence cone about the pole.

[a] Palaeomagnetic sites not used in calculation of the site mean ($\alpha_{95} \geq 15°$).

Oligocene times. As stated above, however, LCF pole may well represent a virtual geomagnetic pole and we are aware that there is not strong evidence for the Paleogene age of this magnetisation. Nevertheless, all above-discussed possibilities demonstrate that palaeomagnetic data from La Chonta Formation present no indication of local rotation of the area during the Andean orogeny (Table 2).

6. Early Permian pole position for Gondwana

As the quality of the Gondwana dataset has been discussed as a possible cause of the misfit of the Permian–Triassic pole for Gondwana with that for Laurasia when adopting the Pangaea A reconstruction, we intend to use the most reliable published results for calculating a mean pole for Gondwana in this paper. We make use of the proposed high quality palaeomagnetic results Early Permian selected by Bachtadse et al. (2002) for this purpose. In addition, we inspected the online Global Palaeomagnetic Database (GPMDB version 4.6, January 2005) to bring the list of reliable Early Permian pole for Gondwana up to date.

According to Bachtadse et al. (2002), who examined the results in the palaeomagnetic database released in June 2000 (GPMDB 3.5), only 11 pole positions out of the published 45 poles for Gondwana are of high quality. At this time, reliable palaeomagnetic results

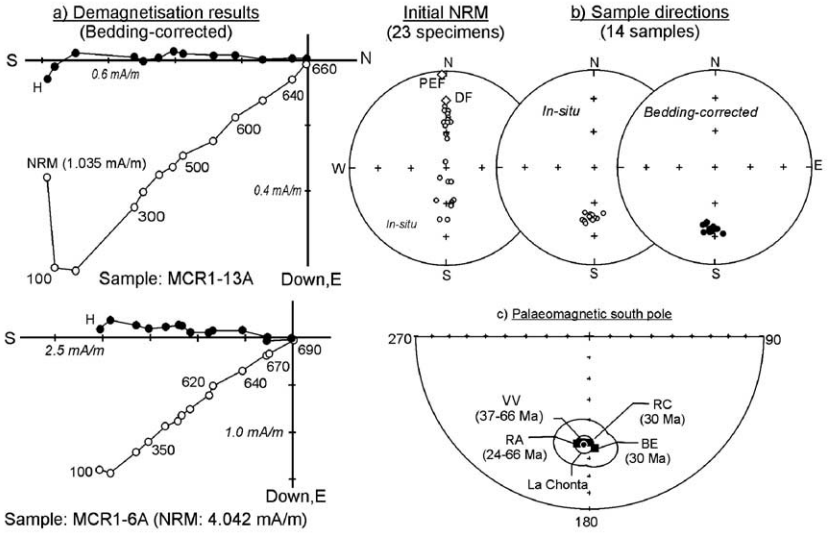


Fig. 6. Representative examples of stepwise demagnetisation results from the Late Cretaceous La Chonta Formation. (a) Orthogonal projection show stable directions well defined between 300 °C and 690 °C. (b) Equal area stereographic projection of the isolated individual sample directions ($n=14$). (c) Equal area projection of the palaeomagnetic south pole from La Chonta Formation compared with the Eocene and Oligocene reference poles for the stable South American plate. Numbers between brackets represent mean age of poles. Reference poles are from Roperch and Carlier (1992, RC), Van der Voo (1993, VV), Randall (1998, RA) and Beck (1998, BE).

interpreted by the original authors to be of primary origin and selected using the reliability criteria of Van der Voo (1990) include six studies from Africa, three studies from Australia and two studies from South America. These authors considered as a principal critical selection the rock age (well determined within half a period), and at least two of the criteria which require sufficient number of samples and adequate statistics, vector analysis of demagnetisation data or positive field tests. Our inspection of GPMDB database version 4.6 released in 2005, which includes palaeomagnetic data published up to December 2004, yields some new poles from South America and Australia. Whereas, no Early Permian palaeomagnetic data from Africa has been published since 2002. For the Australian database, new results reported by Clark and Lackie (2003) supersede the Mount Leyshon Intrusive and Tucker Igneous Complex palaeopoles from North Queensland previously published by Clark (1996). For South America, new Early Permian pole positions from Argentina are presented by Tomezoli (2001) and Geuna and Escosteguy (2004). The Tomezzoli's pole from Tuna Formation (Tuna II: $\lambda=74.1°$S, $\phi=25.9$ °E, $A_{95}=4.8°$) is supported by a positive fold test after McFadden (1990) at 90% unfolding. The author attributed a Late Early Permian to Early Late Permian age (Approximately 254–280 Ma) younger than the previously published Tuna I pole (Tomezzoli and Vilas, 1999), which is considered as having an age interval between 276 Ma and 299 Ma. Since these accredited ages embrace the Early Permian time interval considered in this paper, we include both poles from Tuna Formation in our selected pole for Gondwana. On the other hand, the new Late Carboniferous–Early Permian pole positions reported

Table 2
Selected Early Permian palaeomagnetic data from South America (see text)

	Rock unit	Age (Ma)	λ_{am} (°S)	ϕ_{am} (°E)	λ_{af} (°S)	ϕ_{af} (°E)	α_{95} (°)	Tests	References
TAM	Tambillos Formation	271–285	78.9	319.6	46.5	063.6	5.2	Fo	(1)
TUNI	Tunas Formation, synfold	254–280	63.0	013.9	24.6	068.5	4.8	F+	(2)
TUNII	Tunas Formation, synfold	276–299	74.1	025.9	34.4	076.1	5.2	F+	(3)
COP	Copacabana Group	269–290	68.2	321.3	40.6	051.2	5.2	F+	(4)
	Mean Early Permian pole	254–299	73.3	353.5	36.8	065.3	14.3	$K=42.0$	

λ_{am} (°S)/ϕ_{am} (°S) are pole latitude and longitude in South American coordinates; λ_{af} (°S)/ϕ_{af} (°S), pole positions in African coordinates following the reconstruction parameters of Lawver and Scotese (1987). References are: (1) Rapalini and Vilas (1991), (2) Tomezzoli and Vilas (1999), (3) Tomezzoli (2001), (4) this study. F+: positive fold test; Fo: indeterminate fold test.

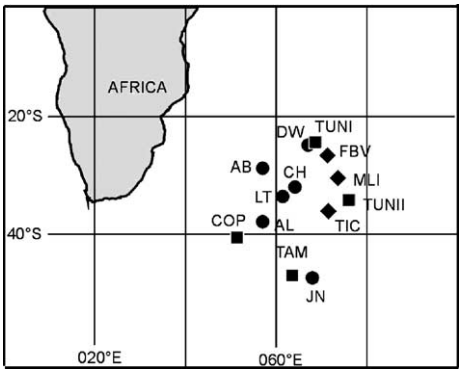


Fig. 7. Projection of the selected palaeomagnetic south pole positions for Gondwana (in African coordinates) used in this paper and listed in Table 3. Circles, squares and diamond represent pole positions from Africa, South America and Australia, respectively. South American and Australia poles are transferred in African coordinates according to the reconstruction parameters of Lawver and Scotese (1987) listed in Table 4.

by Geuna and Escosteguy (2004) from the Paganzo Basin are not consistent between each other and with the previously published data. Since the discrepancy of the Paganzo poles is attributed either to an incorrect age assignment or tectonic rotations related to the Permian San Rafael Orogeny (Geuna and Escosteguy, 2004), we decide to exclude all poles from this basin from our selection. Consequently, among all previously published data for the South American plate, the palaeomagnetic results from Rapalini and Vilas (1991), Tomezzoli and Vilas (1999), and Tomezzoli (2001) are considered in this paper as being reliable and likely reflect the direction of the Earth's magnetic field during the Early Permian times. However, the pole positions from these studies and the new pole reported in this paper are scattered, apparently reflecting the dispersed distribution of all Permian poles for Gondwana (Fig. 7). The cause of this discrepancy is unclear and it cannot be attributed to a single factor since errors in palaeopole positions due to tectonic reasons, inclination shallowing in sediments or existence of non-dipole fields are not obvious.

Because palaeogeographic fits between the Gondwana continents may also cause inconsistency between palaeomagnetic poles, we have tested the reconstruction parameters commonly used for transferring palaeomagnetic data from South America and Australia into African (Gondwana) coordinates. The "great circle distances" (angular separation in degrees) between the mean African pole, used as reference, and the mean Australia poles on one hand, and the South American pole on the other hand were calculated to measure the goodness of continental fits proposed by Lawver and Scotese (1987) and Lottes and Rowley (1990), respectively. The result shown in Table 3 suggests that the least GCD (best fit) for both Africa–Australia (3.7°) and Africa–South America (3.3°) fits are obtained by using the reconstruction of Lawver and Scotese (1987). Hence, the reconstruction parameters proposed by these authors were used in this paper when transferring palaeomagnetic poles from Australia and South America into African coordinates. Consequently, the selected high quality poles listed in Table 4 and the Copacabana pole yields a combined Early Permian pole for Gondwana situated at $\lambda = 34.4°$S, $\phi = 065.6°$E ($A_{95} = 4.9°$, $K = 73.6$, $N = 13$ studies) in African coordinates. It is worth to notice that the mean South American poles [$\lambda = 36.9°$S, $\phi = 065.3°$E ($A_{95} = 14.3°$, $K = 42.0$, $N = 4$ studies)] plots within the interval of confidence for the Early Permian pole position calculated exclusively from African and Australian data [$\lambda = 33.3°$S, $\phi = 065.7°$E ($A_{95} = 5.4°$, $K = 92.6$, $N = 9$ studies)]. The mean Early Permian pole position from South America is, therefore, statistically indistinguishable at the 95%

Table 3
Position of the mean Early Permian poles from South America and Australia transferred in African coordinates according to the reconstruction parameters of Lawver and Scotese (1987) and Lottes and Rowley (1990), and the calculated great circle distance (GCD in degrees) between African–South American and African–Australian mean poles

	Local coordinates			Lawver and Scotese (1987)			Lottes and Rowley (1990)		
	λ (°S)	ϕ (°E)	A_{95} (°)	λ (°S)	ϕ (°E)	GCD (°)	λ (°S)	ϕ (°E)	GCD (°)
Africa	–	–	–	34.2	062.9	(A_{95}=7.1°)	28.8	065.5	(A_{95}=6.0°)
Australia	44.4	136.9	7.4	31.3	072.2	3.7	28.7	073.4	6.9
South America	73.3	353.5	14.3	36.8	065.3	3.3	39.6	065.3	10.8
Reconstruction parameters									
South America–Africa				45.5°N/327.8°E/+58.2°			46.82°N/329.46°E/+55.8°		
Australia–Africa	°			(1) 1.58°S/039.02°E/−31.29			29.26°S/302.81°E/+54.02°		
				(2) 7.78°S/328.58°E/+58.0°					

In Lottes and Rowley (1990), the mean pole for Africa (in Central African coordinates) were calculated considering relative rotations of northwestern Africa (7.8°) and northeastern Africa (6.3°). Conversely, Africa is considered as a single plate in Lawver and Scotese (1987) reconstructions (1) and (2) are reconstruction parameters for Australia/East Antarctica and East Antarctica/Africa, respectively.

Table 4
Previously published palaeopoles for Gondwana derived from high quality palaeomagnetic data with well-determined Early Permian age

	Rock unit	Age (Ma)	λ (°S)	ϕ (°E)	α_{95} (°)	Tests	References
Africa							
CH	Chougrane red beds	273 ± 18	32.2	064.1	4.7	F+	Daly and Pozzi (1976)
LT	Lower Tiguentourine Fm.	290 ± 10	33.8	061.4	4.1	–	Derder et al. (1994)
AB	Abdala Formation, lower unit	285 ± 14	29.0	057.0	3.0	–	Merabet et al. (1998)
AL	Upper El Adeb Larache Fm.	295 ± 11	38.0	057.0	2.0	–	Henry et al. (1992)
DW	Dwyka Fm., combined data	282 ± 21	25.0	067.0	12.0	F+	Opdyke et al. (2001)
JN	Jebel Nehoud Ring Complex	280 ± 2	46.9	068.0	5.1	–	Bachtadse et al. (2002)
Australia							
MLI	Mount Leyshon Intrusive	283 ± 4	30.7	073.6	3.0	C+	Clark and Lackie (2003)
FBV	Featherbed Volcanics	293 ± 13	26.8	071.4	5.0	–	Klootwijk et al. (1993)
TIC	Tucker Igneous Complex	287 ± 4	36.2	071.4	3.0	F+	Clark and Lackie (2003)
South America							
TAM	Tambillos Formation	278 ± 7	46.5	063.6	5.2	Fo	Rapalini and Vilas (1991)
TUN I	Tunas Formation, synfold	288 ± 11	24.6	068.5	4.8	F+	Tomezzoli and Vilas (1999)
TUNII	Tunas Formation	267 ± 13	34.4	076.1	5.2	F+	Tomezzoli (2001)
COP	Copacabana Group	280 ± 11	40.6	051.2	5.2	F+	This study
Mean pole for Gondwana		280 ± 25	34.4	065.6	A_{95}=4.9°	K=73.6	N=13 studies

λ (°S) and ϕ (°E) are latitude and longitude positions of pole in African coordinates following the reconstruction parameters of Lawver and Scotese (1987). α_{95} (°) and A_{95} (°): statistical parameters according to Fisher (1953). F+: positive fold test; C+: positive contact test.

confidence level from the pole position calculated from high quality results from the rest of Gondwana.

In view of the Pangaea palaeogeography, the presented pole for Gondwana disagrees with the Early Permian pole for Laurasia and results in an unacceptable continental overlap (1200 ± 500 km) when adopting the Pangaea A configuration. Conversely, palaeoreconstruction of Pangaea based on this pole advocates the Pangaea B configuration, in which Gondwana is placed further east relative to its position in Pangaea A (Fig. 8).

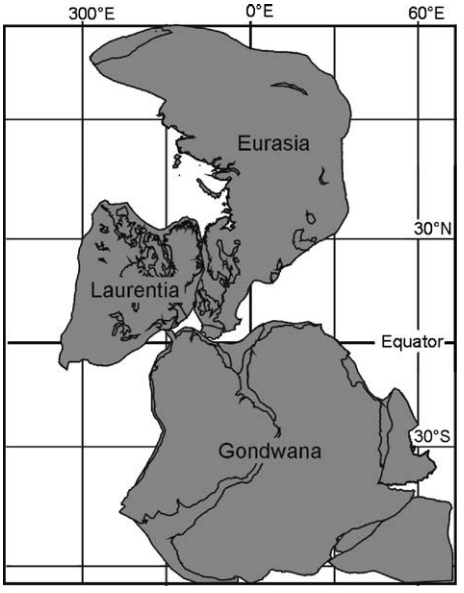


Fig. 8. Early Permian palaeogeographic reconstruction of Pangaea based on the combined palaeomagnetic data from Gondwana (λ=34.4°S, ϕ=064.7°E, A_{95}=5.1°, N=12 studies) and the Early Permian palaeomagnetic poles for Laurasia (λ=46.0°S, ϕ=304.0°E) from Van der Voo (1993). This reconstruction is similar to the proposed Pangaea B configuration. Adoption of the Pangaea type A clearly provides overlap between Gondwana and Laurentia.

7. Conclusion

The new palaeomagnetic result from Copacabana Group combined with the coeval previously published and high quality data from South America, Africa and Australia yield a mean Early Permian pole for Gondwana that disagrees with the coeval pole for Laurasia when assuming the Pangaea type A configuration. The mean pole position for Gondwana proposed in this study is computed from the highest quality data with well-constrained ages and from a broad paleolatitudinal band (3°S to 67°S in Pangaea reconstruction). This new pole position is similar to and further refines the pole for Gondwana calculated by Bachtadse et al. (2002). Considering the mean pole for Gondwana reported in this paper (Pangaea B in Early Permian) and the palaeomagnetic results from southern Peru presented in Rakotosolofo et al. (in preparation), which suggest a Pangaea type A2 in Early Triassic, the palaeomagnetic data are increasingly supporting the hypothesis that the Pangaean palaeogeography evolved from type B to A2 during the Permian–Triassic times as previously suggested by Muttoni et al. (2003). Moreover, recently

published geological and palaeontological evidence reported in Berthelin et al. (2003) and Vai (2003) also support this model.

Hence, we believe that the disagreement of the palaeomagnetic data with the Pangaea A in Permian times is not an artefact of the poor quality results from Gondwana. Palaeolatitudinal errors due to inclination shallowing (Van der Voo, 1993; Rochette and Vandamme, 2001) cannot be ruled out completely, as selected data used in this paper include poles derived from sediments that are not exempted from possible shallowing of the palaeodirections. However, as demonstrated in Muttoni et al. (2003), based on an Early Permian pole position for Gondwana computed exclusively from volcanic rocks from southern Alpine, inclination error in sediments does not explain the problem of Pangaea palaeogeography. Lastly, palaeomagnetic data presented in this paper are not relevant for discussion of the palaeolatitudinal biases due to significant non-dipole field contamination of the Earth's geocentric axial dipole field (Van der Voo and Torsvik, 2001; Torsvik and Van der Voo, 2002). We, however, uphold the basic principle of the palaeogeographic reconstruction based on palaeomagnetism (GAD hypothesis) assuming that the time-averaged geomagnetic field was geocentric axial dipolar.

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